

2. Let $(\Omega, \mathcal{A}, P)$ be a probability space

(a) Show that for any two events $A, B \in \mathcal{A}$,
 $K(A, B) = P(A \cap B)$ is pos. definite.

Choose $(\alpha_1, \dots, \alpha_n) \in \mathbb{R}^n$. Then

$$\begin{aligned} \sum_{ij}^n \alpha_i \alpha_j K(x_i, x_j) &= \sum_{ij}^n \alpha_i \alpha_j P(A_i \cap A_j) = \\ \sum_{ij}^n \alpha_i \alpha_j E(\mathbb{I}_{A_i} \mathbb{I}_{A_j}) &= E\left(\left(\sum_i \alpha_i \mathbb{I}_{A_i}\right) \left(\sum_j \alpha_j \mathbb{I}_{A_j}\right)\right) = \\ E\left(\left(\sum_k \alpha_k \mathbb{I}_{A_k}\right)^2\right) &\geq 0 \end{aligned}$$

(b) Show that $K(A, B) = P(A \cap B) - P(A)P(B)$ is also pos. definite.

Let $A = (A_1, A_2, \dots, A_n)$ be a vector of events.

$$\begin{aligned} \sum_{ij} \alpha_i \alpha_j (P(A \cap B) - P(A)P(B)) &= \sum_{ij} \alpha_i \alpha_j E(\mathbb{I}_{A \cup B}) = \\ \sum_{ij} \alpha_i \alpha_j (E(\mathbb{I}_{A_i} \mathbb{I}_{A_j}) - E(\mathbb{I}_{A_i})E(\mathbb{I}_{A_j})) &= \\ \sum_{ij} \alpha_i \alpha_j Cov(\mathbb{I}_{A_i}(\omega), \mathbb{I}_{A_j}(\omega)) \end{aligned}$$

where $\mathbb{I}_A(\omega) = 1$ if $\omega \in A$ and $\mathbb{I}_A(\omega) = 0$, otherwise. Furthermore

$$\begin{aligned} \sum_{ij} \alpha_i \alpha_j Cov(\mathbb{I}_{A_i}(\omega), \mathbb{I}_{A_j}(\omega)) &= \\ \sum \alpha_i^2 Var(\mathbb{I}_{A_i}(\omega)) + 2 \sum_{i,j|i < j} \alpha_i \alpha_j Cov(\mathbb{I}_{A_i}(\omega), \mathbb{I}_{A_j}(\omega)) &= \\ \sum_i^n Var(\mathbb{I}_{A_i}(\omega)) &\geq 0. \end{aligned}$$